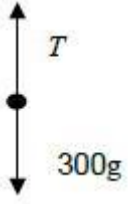
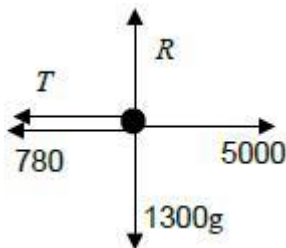


Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Forms an equation for gradient of $CD = \frac{1}{4}$ or $-\frac{1}{4}$ of the form difference in y over difference in x (or vice versa = 4 or -4)	AO3.1a	M1	$\frac{d-2}{6-c} = \frac{1}{4}$ $4d - 8 = 6 - c$ $c + 4d = 14$ $\frac{2+d}{2} + 4\left(\frac{c+6}{2}\right) = 11$ $4c + d = -4$ $c = -2 \qquad d = 4$
Obtains a correct equation for c & d	AO1.1b	A1	
Forms an equation for the mid-point of CD lying on $y + 4x = 11$	AO3.1a	M1	
Obtains correct equation for c & d (any correct form)	AO1.1b	A1	
Solves for c and d CAO	AO1.1b	A1	
			Total 5 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Draws correct force diagram for crate from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	
(i)				
(ii)	Draws correct force diagram for van from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	

(b)	Applies Newton's 2nd Law ($F = ma$) to the crate	AO3.4	M1	For crate $T - 300g = 300a$
	Applies Newton's 2nd Law ($F = ma$) to the van ($F = ma$ 'round the corner' scores 0)	AO3.4	M1	For van $5000 - T - 780 = 1300a$ $4220 - T = 1300a$
	Solve their simultaneous equations	AO1.1a	M1	$4220 - 300g = 1600a$
	Finds the value of a correctly AG	AO1.1b	A1	$a = 1280 \div 1600 = 0.80 \text{ m s}^{-2}$ (AG)
(c)	Uses $a = 0.80$ in either of their two equations in (b)	AO3.4	M1	$T = 300 \times 0.80 + 300g$
	Finds the correct value for T (condone omission of units) Possibly done in (b)	AO1.1b	A1	$= 3180$ $= 3200 \text{ (N) (2 sf)}$
(d)	Explains that the model could be refined by including air resistance	AO3.5c	E1	Resistance will increase with speed
Total 9 marks				

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Recalls $A = \frac{1}{2}r^2\theta$ or $l = r\theta$ PI by use in equation	AO1.2	B1	Area of sector gives $\frac{1}{2}r^2\theta = 9$, $\theta = \frac{18}{r^2}$
	Constructs two equations at least one correct	AO1.1a	M1	Perimeter of sector gives $2r + r\theta = 15$
	Eliminates θ FT incorrect equations	AO1.1a	M1	$2r + \frac{18}{r} = 15$
	Constructs a rigorous mathematical argument to show required result, clearly constructing two correct simultaneous equations and eliminating θ AG	AO2.1	R1	$2r^2 + 18 = 15r$ $2r^2 - 15r + 18 = 0$ (AG)

(b)	Solves a quadratic equation and finds two values of θ	AO3.1a	M1	$r = \frac{3}{2}, r = 6$
	Finds two correct values of r	AO1.1b	B1	$r = 6 \Rightarrow \theta = \frac{1}{2}$
	Finds both values of θ	AO1.1b	A1	$r = \frac{3}{2} \Rightarrow \theta = 8$
	Gives a valid reason for rejecting one of 'their' values	AO2.4	R1	$8 > 2\pi \therefore \theta \neq 8$ so only one possible value of θ
Total 8 marks				

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies and clearly defines variables.	AO3.1b	B1	Let C = total cost x = length of base edges h = length of height
	Models the cost with an expression of the form $ax^2 + bxh$	AO3.3	M1	$C = 15x^2 + 32xh$
	Eliminates either variable, using volume equation, to form a model for the cost in one variable.	AO3.3	M1	$x^2h = 60$ $h = \frac{60}{x^2}$
	Obtains a correct equation to model cost in one variable	AO3.1b	A1	$C = 15x^2 + \frac{1920}{x}$ (*)
	Uses their model to find minimum. (at least one term correctly differentiated and equated to zero)	AO3.4	M1	Differentiating $30x - \frac{1920}{x^2} = 0$ $x^3 = 64$ $x = 4$ $h = \frac{60}{4^2} = 3.75$
	Obtains correct equation	AO1.1b	A1	Height of tank is 3.75 m $\frac{d^2C}{dx^2} = 30 + \frac{3840}{x^3}$
	Obtains correct value for h with correct units in context Award FT from correct substitution into incorrect equation for h but only if all three M1 marks have been awarded, must have	AO3.2a	A1F	$x = 4 \Rightarrow \frac{d^2C}{dx^2} > 0$ therefore minimum ALT from (*) onwards $C = 900h^{-1} + 32\sqrt{60}h^{\frac{1}{2}}$ $0 = -900h^{-2} + 16\sqrt{60}h^{-\frac{1}{2}}$

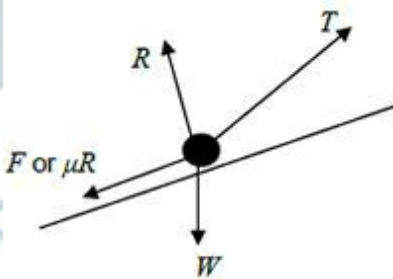
correct units.			Height of tank is 3.75 m
Performs a correct test of 'their' solution: uses the second derivative of 'their' expression for C in terms of x or h to justify that a minimum value for h has been found OE (Second derivative > 0 or test gradient/values either side)	AO2.4	R1	$\frac{d^2C}{dh^2} = 1800h^{-3} - 8\sqrt{60}h^{-\frac{3}{2}}$ $h = 3.75 \Rightarrow \frac{d^2C}{dh^2} = 25.6 > 0$ $\therefore \text{minimum}$

(b)	Explains that two of the sides of the tank will need to have length $x \pm 0.05$ in order to join them	AO3.5c	E1	The sides will need to overlap to be joined, so two of the side lengths will need to be $x + 0.05$
(i)				
(ii)	Explains that the refinement is relatively small and unlikely to have a significant effect on the result.	AO3.5a	R1	The minimum cost is likely to increase slightly, but relative to the size of the tank, an extra 5cm is unlikely to make a significant difference.
Total 10 marks				

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States correct expression for a	AO1.1b	B1	$a = \frac{V - U}{T}$
(b)	Rearranges to make T the subject of the formula	AO2.1	R1	$T = \frac{V - U}{a}$
	Uses given expression for S and attempts to eliminate T	AO2.1	R1	$S = \frac{1}{2}(U + V) \times \frac{V - U}{a}$ $2as = (U + V)(V - U)$
	Completes argument to reach required result AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$V^2 = U^2 + 2aS$ (AG)
Total 4 marks				

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves horizontally and vertically to obtain expressions for F and R (allow consistent mixing of sin and cos)	AO3.4	M1	Resolving vertically $R = 8 \times 9.8 - 50\sin 40^\circ$ Resolving horizontally $F = 50\cos 40^\circ$
	Obtains correct expressions for R and F	AO1.1b	A1	
	States friction model $F = \mu R$ with 'their' values for F and R	AO1.2	B1	
	Completes a rigorous argument that results with correct μ AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$F = \mu R$ $50\cos 40^\circ = \mu(8 \times 9.8 - 50\sin 40^\circ)$ $\mu = \frac{50\cos 40^\circ}{8 \times 9.8 - 50\sin 40^\circ}$ $\mu = 0.8279660445 = 0.83$ (2sf) AG
(b)	Draws correct diagram with exactly four forces showing arrow heads and labels Can use Mg or $8g$ or 78.4 for W	AO3.3	B1	
(ii)	Resolves perpendicular to the plane resulting in a three term equation containing; R , $8g\cos 5^\circ$ and $T\sin 40^\circ$ (or $T\cos 50^\circ$) OR Resolves horizontally and vertically to obtain equations of motion in horizontal and vertical directions	AO3.1b	M1	$R = 8 \times 9.8\cos 5^\circ - T\sin 40^\circ$ $= 78.1 - T\sin 40^\circ$
	Obtains correct expression	AO1.1b	A1	$T\cos 40^\circ - 8 \times 9.8\sin 5^\circ - F$

for R OR Obtains correct horizontal and vertical equations			$= 8 \times 3$ $T \cos 40^\circ - 8 \times 9.8 \sin 5^\circ - 0.827 \dots \times (8 \times 9.8 \cos 5^\circ - T \sin 40^\circ) = 24$
Forms a four term equation of motion parallel to the plane with correct terms (allow sign errors) OR Eliminates R to solve for T	AO3.1b	M1	$T = \frac{24 + 8 \times 9.8 \sin 5^\circ + 0.827 \dots \times 8 \times 9.8 \cos 5^\circ}{\cos 40^\circ + 0.827 \dots \times \sin 40^\circ}$ $T = 73.55939193 = 74 \text{ (2 sf)}$
Obtains correct equation of motion OR Obtains correct expression(s) without R	AO1.1b	A1	
Uses the friction model in the four term equation of motion where R is in the form $a - bT$ and a and b are positive constants OR Uses the friction model in the horizontal and vertical equations Dependent on previous M1	AO3.1b	dM1	
Solves for T	AO1.1b	A1	
Obtains correct value of T with 2 sf accuracy. FT incorrect value found for T provided both M1 marks and dM1 mark have been awarded	AO3.2a	A1F	
			Total 12 marks

Q7.

- (a) (i) $21 + 5k = 1$
 condone $3 \times 7 + 5k = 1$
 $\Rightarrow k = -4$
 AG *condone* $y = -4$

B1

(ii) $(x =) 2$

B1

$(y =) -1$

midpoint coords are (2, -1)

B1

2

(b) $y = \frac{1}{5} - \frac{3}{5}x$

obtaining $y = a \pm \frac{3}{5}x$ or $\frac{\Delta y}{\Delta x} = \frac{-4-2}{7--3}$ or $\frac{-1-2}{2--3}$ or $\frac{-4--1}{7-2}$

*condone **one** sign error in expression*

M1

$(\text{Gradient } AB =) -\frac{3}{5}$

allow -0.6, $-\frac{6}{10}$ etc for A1 & condone error in rearranging if gradient is correct.

A1

2

(c) Perp grad = $\frac{5}{3}$

-1 / "their" grad AB

M1

$$\left. \begin{array}{l} y - 2 = \frac{5}{3}(x + 3) \\ \text{or } y = \frac{5}{3}x + c, \quad c = 7 \quad \text{etc} \end{array} \right\}$$

correct equation in any form (must simplify $x - -3$ to $x + 3$ or c to a single term equivalent to 7)

A1

$5x - 3y + 21 = 0$

or any multiple of this with integer coefficients – terms in any order but all terms on one side of equation

A1

3

(d) $3x + 5y = 1$ and $5x + 8y = 4 \Rightarrow Px = Q$ or $Ry = S$
*must use **correct** pair of equations **and** attempt to*

eliminate y (or x) (generous)

M1

$$x = 12$$

A1

$$y = -7$$

$$(12, -7)$$

A1

3

[11]

Q8.

(a) $2x^2 - x - 1 = 2kx - 3k$
 $2x^2 - x - 1 - 2kx + 3k = 0$ OE
 $\Rightarrow 2x^2 - (2k + 1)x + 3k - 1 = 0$

equated and multiplied out and all 5 terms on one side and = 0

AG (correct with no trailing = signs etc)

B1

1

(b) (i) $(2k + 1)^2 - 4 \times 2(3k - 1)$
clear attempt at $b^2 - 4ac$

M1

$$(2k + 1)^2 - 4 \times 2(3k - 1) > 0$$

discriminant > 0 which must appear before the printed answer

B1

$$4k^2 + 4k + 1 - 24k + 8 > 0 \Rightarrow 4k^2 - 20k + 9 > 0$$

AG (all working correct with no missing brackets etc)

A1cso

3

(ii) $4k^2 - 20k + 9 = (2k - 9)(2k - 1)$

correct factors or correct use of formula as far as

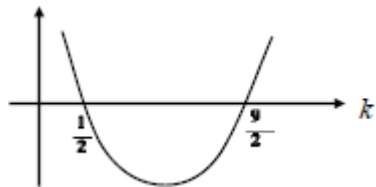
$$\frac{20 \pm \sqrt{256}}{8}$$

M1

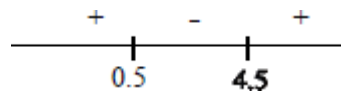
critical values are $\frac{1}{2}$ and $\frac{9}{2}$

condone $\frac{4}{8}$, $\frac{36}{8}$ etc here but must combine sums of fractions

A1



sketch or sign diagram including values



M1

$$k < \frac{1}{2}, \quad k > \frac{9}{2}$$

fractions must be simplified condone use of OR but not AND

A1

Take their final line as their answer.

4

[8]

Q9.

(a) $3p - 4(p + 2) + 5 = 0$

condone omission of brackets or one sign error

M1

$$(\Rightarrow p =) -3$$

A1

2

(b) $y = \frac{3}{4}x + \frac{5}{4}$

rearranging into form $y = \pm \frac{3}{4}x + c$

M1

(gradient $AB =$) $\frac{3}{4}$

condone slips in rearranging if gradient is correct.

A1

2

(c) $\text{gradient } AC = \frac{k-2}{-5-1}$
 or $\frac{2-k}{1-5}$ (condone one sign error)

M1

“their” $\frac{(k-2)}{-6} \times \frac{3}{4} = -1$ OE
 product of grads = -1 in terms of k

m1

$(\Rightarrow k =) 10$

A1

3

(d) $3x - 4y + 5 = 0$ and $2x - 5y = 6$
 $\Rightarrow P \ x = Q$ or $R \ y = S$
 *must use “correct” pair of equations **and** attempt to eliminate y (or x) (generous)*

M1

$x = -7$

A1

$y = -4$

A1

$(-7, -4)$

3

[10]

Q10.

(a) $\frac{dy}{dx} = 5x^4 - 4x$
 one of these terms correct

M1

all correct (no + c etc)

A1

$$= 5(-1)^4 - 4(-1)) = 9$$

A1

Tangent has equation $y = \text{'their 9'}x + c$

and $6 = \text{'their 9'}(-1) + c \Rightarrow = \dots$

*tangent using 'their' gradient, and attempt to find c
using $x = -1$ and $y = 6$*

m1

$$\Rightarrow y = 9x + 15$$

equation must be seen in this form

A1

5

(b) (i) When $x = 2$, $y = 2^5 - 2 \times 2^2 + 9 = 32 - 8 + 9 = 33$

*be convinced that they are using **curve** equation*

$(k =) 33$

NMS $k = 33$ scores B0

B1

1

(ii) When $x = 2$, $y = 9 \times 2 + 15 = 33$ so lies on tangent

*be convinced that they are using **tangent** equation
and have statement*

B1

1

(c) (i) $\frac{x^6}{6} - \frac{2x^3}{3} + 9x$

one of these terms correct

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M1

another term correct

A1

all correct (may have +c)

A1

$$\left[\frac{2^6}{6} - \frac{2 \times 2^3}{3} + 9 \times 2 \right] - \left[\frac{(-1)^6}{6} - \frac{2 \times (-1)^3}{3} + 9 \times (-1) \right]$$

$$\left[\frac{64}{6} - \frac{16}{3} + 18 \right] - \left[\frac{1}{6} + \frac{2}{3} - 9 \right]$$

= 31.5

$F(2) - F(-1)$ unsimplified FT "their terms" from
 integration = $\frac{70}{3} - \left(-\frac{49}{6}\right)$

m1

$\left(\text{or } \frac{189}{6} \text{ etc}\right)$

condone single fraction

A1

5

(ii) Area of trapezium = $\frac{1}{2} \times 3 \times (6 + \text{'their' } k)$
 = 58.5 when $k = 33$

B1✓

Shaded area = **Trapezium** – 'their' (c)(i) value

M1

= 27

OE eg $\frac{162}{6}$

A1

3

[15]

Q11.

(a) $(x + 7)^2 + (y - 5)^2$
 one term correct ; condone $(x - -7)^2$

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M1

both terms correct with squares
 and plus sign between terms

A1

$(x + 7)^2 + (y - 5)^2 = 5^2$
 condone 25 for 5^2

A1cao

(b) (i) $C(-7, 5)$
 correct or FT 'their' circle equation

B1✓

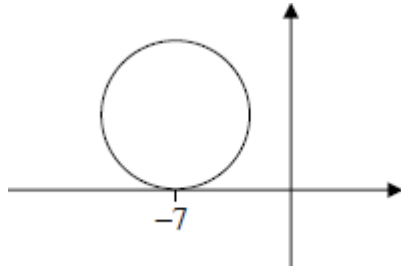
(ii) $r = 5$

correct or FT 'their' $r^2 > 0$

condone $\sqrt{25}$ etc but not $\pm \sqrt{25}$

B1✓

(c) must draw axes



freehand circle with C correct or FT 'their C ' for quadrant of centre

M1

circle touching x -axis at -7 with -7 marked (need not show 5 on y -axis) but circle must not touch y -axis

A1

(d) (i) $x^2 + (kx + 6)^2 + 14x - 10(kx + 6) + 49 = 0$

clear attempt to sub $y = kx + 6$ into original or 'their' circle equation ...

$$x^2 + k^2x^2 + 12kx + 36 + 14x - 10kx - 60 + 49 = 0$$

...and attempt to multiply out

M1

$$(1 + k^2)x^2 + 2kx + 14x + 25 = 0$$

$$\Rightarrow (k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$$

AG condone $x^2(1 + k^2) + 2x(7 + k) + \dots$ etc

A1cso

(ii) Equal roots ' $b^2 - 4ac = 0$ '

allow statement alone if discriminant in terms of k attempted

B1

$$[2(k + 7)]^2 - 4 \times 25(k^2 + 1)$$

discriminant (condone one slip)

M1

$$4\{k^2 + 14k + 49 - 25k^2 - 25\} = 0$$

$$-24k^2 + 14k + 24 = 0$$

$$\Rightarrow 12k^2 - 7k - 12 = 0$$

AG all working correct
but = 0 must appear before last line

A1

3

(iii) $(4k + 3)(3k - 4)$

correct factors or correct use of formula as far as

$$k = \frac{7 \pm \sqrt{49 + 576}}{24}$$

M1

$$\Rightarrow k = -\frac{3}{4}, k = \frac{4}{3} \quad \text{OE}$$

are values of k for which line is a tangent

A1

2

[14]

Q12.

(a) $\frac{dy}{dx} = 12 - 5x^{\frac{2}{3}}$
 $kx^{\frac{2}{3}}$ term.


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ACF

M1

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2

(b) (i) When $x = 0$, $\frac{dy}{dx} = 12$

Ft on c's y' evaluated correctly at $x = 0$

B1F

Eqn of tangent at O is $y = 12x$

OE Ft on c's value for $y'(0)$ provided $y'(0) > 0$.

B1F

2

(ii) When $x = 8$, $\frac{dy}{dx} = 12 - 5 \times (8)^{\frac{2}{3}}$

Attempt to find $\frac{dy}{dx}$ when $x = 8$

M1

Equation of tangent at $(8, 0)$ is $y - 0 = y'(8)[x - 8]$

$y = y'(8)[x - 8]$ OE

M1

$y = -8(x - 8) \Rightarrow y + 8x = 64$

CSO AG

A1

3

(c) $\int \left(12x - 3x^{\frac{5}{3}} \right) dx = \frac{12x^2}{2} - \frac{3x^{\frac{8}{3}}}{\frac{8}{3}} (+c)$

$kx^{\frac{5}{3}+1}$ term after integrating, condone k left unsimplified for this M mark.

$= 6x^2 - \frac{9}{8}x^{\frac{8}{3}} (+c)$

For $6x^2$ OE eg $(12x^2 / 2)$

B1

For $-\frac{9}{8}x^{\frac{8}{3}}$ OE

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A1

3

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(d) Area bounded by curve and x -axis

$= \int_0^8 \left(12x - 3x^{\frac{5}{3}} \right) dx = 6 \times 8^2 - \frac{9}{8} \times (8)^{\frac{8}{3}}$

$\pm F(8) \{- F(0)\}$ PI following integration

M1

$= 384 - 288 = 96$

PI by correct final answer if evaluation not seen here

A1

At P , $12x + 8x = 64$

Solving $y + 8x = 64$ and c's $y = kx$, $k > 0$, down to an eqn in one variable...

$$[y + 2y / 3 = 64]$$

M1

$$(x_p = 3.2) \quad y_p = 38.4$$

For $y_p 38.4$ OE [If using integration to find area of triangle, award A1 if both ' $x_p = 3.2$ ' and correct integration of correct eqns of the 2 lines]

A1

$$\text{Area of triangle } OPA = \frac{1}{2} \times 8 \times y_p$$

OE Need perpendicular ht to be linked to $y_p > 0$.

M1

$$\text{Area of shaded region} = \text{Area } \Delta OPA - \int_0^8 \left(12x - 3x^{\frac{5}{3}} \right) dx$$

M0 if evaluated to a value < 0

M1

$$= 153.6 - 96 = 57.6$$

OE eg $288 / 5$

A1

7

[17]

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Q13.

$$(a) \quad (i) \quad y = \frac{4}{3}x - \frac{7}{3}$$

$$y = \pm \frac{4}{3}x + k \text{ or } \frac{\Delta y}{\Delta x} \text{ with 2 correct points}$$

M1

$$\Rightarrow \text{grad } AB = \frac{4}{3}$$

condone slip in rearranging if gradient is correct; condone 1.33 or better

A1

2

- (ii) $y = \text{'their grad'} \ x + c$
and attempt to use $x = 3, y = -5$
or $y - -5 = \text{'their grad } AB' (x - 3)$
or $4x - 3y = k$ and attempt to find k using
 $x = 3$ and $y = -5$

M1

$$\left. \begin{array}{l} y + 5 = \frac{4}{3}(x - 3) \\ \text{or } y = \frac{4}{3}x - \frac{27}{3} \end{array} \right\}$$

correct equation in any form but must
simplify -- to +

A1

$$4x - 3y = 27$$

integer coefficients in required form

$$\text{eg } -8x + 6y = -54$$

A1

3

(b) $4x - 3y = 7$ and $3x - 2y = 4$

$$\Rightarrow 8x - 9x = 14 - 12 \text{ etc}$$

must use **correct pair** of equations and
attempt to eliminate x or y (generous)

M1

$$x = -2$$

A1

$$y = -5$$

$$\text{or } D(-2, -5)$$

A1

3

(c) $4(k - 2) - 3(2k - 3) = 7$

$$\text{sub } x = k - 2, y = 2k - 3 \text{ into } 4x - 3y = 7$$

$$4k - 8 - 6k + 9 = 7$$

and attempt to multiply out with all k
terms on one side (condone one slip)

M1

$$\Rightarrow k = -3$$

A1

2

[10]

Q14.

(a) (i) $\left(x - \frac{3}{2}\right)^2$
 or $p = 1.5$ stated

M1

$$\left(x - \frac{3}{2}\right)^2 + \frac{11}{4}$$

$$(x - 1.5)^2 + 2.75$$

A1

Mark their final line as their answer

2

(ii) $x = \frac{3}{2}$
 correct or FT their " $x = p$ "

B1 ✓

1

(b) (i) $x^2 - 3x + 5 = x + 5 \Rightarrow x^2 = 4x$
 eliminating x or y and collecting like terms
 (condone **one** slip)
 or $(y - 5)^2 - 3(y - 5) + 5 = y \Rightarrow y^2 - 14y + 45 = 0$

M1

$$(x \neq 0) \quad \Rightarrow x = 4$$

A1

$$y = 9$$

A1

3

(ii) $\frac{x^3}{3} - \frac{3x^2}{2} + 5x (+ c)$
 one of these terms correct

M1

another term correct

A1

all correct (need not have + c)

A1

3

(iii) $\left[\right]_0^4 = \frac{4^3}{3} - 3 \times \frac{4^2}{2} + 5 \times 4$

must have earned M1 in part(b)(ii)

F(their x_B){-F(0)} "correctly sub'd"

M1

$= 17\frac{1}{3}$

$\left(\frac{64}{3} - 24 + 20 = \right) \frac{52}{3} \text{ or } \frac{104}{6} \text{ etc}$

condone 17.3 but not $16\frac{4}{3}$ etc

A1

Area trapezium $\frac{1}{2}(x_B)(5 + y_B)$

FT their numerical values of x_B , y_B

$\text{Area} = \frac{1}{2} \times 4 \times 14 (= 28)$

B1 ✓

Area of shaded region $28 - 17\frac{1}{3}$

$= 10\frac{2}{3}$

CSO; $\frac{32}{3}$, accept 10.7 or better

A1

4

[13]

Q15.

(a) (i)

$$\left. \begin{array}{l} \text{(Increasing} \Rightarrow \frac{dy}{dx} > 0 \\ 20x - 6x^2 - 16 > 0 \end{array} \right\} \text{ either}$$

correct interpretation of y increasing

M1

$$\Rightarrow 6x^2 - 20x + 16 < 0 \quad \text{or} \quad (2) \quad (10x - 3x^2 - 8) > 0$$

must see at least one of these steps before final answer for A1

$$\Rightarrow 3x^2 - 10x + 8 < 0$$

CSO **AG** no errors in working

A1

2

(ii) $(3x - 4)(x - 2)$

correct factors or correct use of quadratic equation

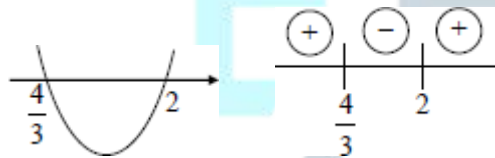
formula as far as $\frac{10 \pm \sqrt{4}}{6}$

M1

CVs are $\frac{4}{3}$ and 2

condone $\frac{8}{6}$ and $\frac{12}{6}$ here but not in final line

A1



sketch or sign diagram

M1

$$\frac{4}{3} < x < 2$$

$$\text{or } 2 > x > \frac{4}{3}$$

$$\text{accept } x < 2 \text{ AND } x > \frac{4}{3}$$

$$\text{but not } x < 2 \text{ OR } x > \frac{4}{3}$$

$$\text{nor } x < 2, x > \frac{4}{3}$$

A1

Mark their final line as their answer

4

(b) (i) $x = 2 ; \left(\frac{dy}{dx} = \right) 40 - 24 - 16$

sub $x = 2$ into $\frac{dy}{dx}$ and simplify terms

M1

$\frac{dy}{dx} = 0 \Rightarrow$ tangent at P is parallel to the x -axis
must be all correct working plus statement

A1

2

(ii) $x = 3 ; \frac{dy}{dx} = 20 \times 3 - 6 \times 3^2 - 16$

must attempt to sub $x = 3$ into $\frac{dy}{dx}$

M1

$(= 60 - 54 - 16) = -10$

A1

Gradient of normal = $\frac{1}{10}$
 $\frac{-1}{\text{"their" } -10}$

A1 ✓

Normal: $(y - -1) = \text{'theirgrad'}(x - 3)$
normal attempted with correct coordinates

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used and gradient obtained from their $\frac{dy}{dx}$ value

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$y + 1 = \frac{1}{10}(x - 3)$

*any correct form, eg $10y = x - 13$ but
 must simplify -- to +*

A1

(Equation of tangent at P is) $y = 3$

B1

$x = 43$

CSO; $\Rightarrow R (43, 3)$

A1

7

[15]

Q16.

(a) (i) $y = \frac{1}{2}(7 - 3x)$

attempt at $y = \dots$

or use of 2 correct points using $\frac{\Delta y}{\Delta x}$

M1

$\Rightarrow \text{gradient} = -\frac{3}{2}$

condone slip in rearranging if gradient is correct

A1

2

- (ii) $y = \text{'their grad'} x + c$
 and substitution of $x = 2, y = -7$
or using $3x + 2y = k$ with $x = 2, y = -7$
and attempt to find k or $y - -7 = \text{'their grad'} (x - 2)$

M1

$y = -\frac{3}{2}x + c, c = -4$

correct equation in any form

$y + 7 = -\frac{3}{2}(x - 2), 3x + 2y + 8 = 0, \text{etc}$

A1

$(x = 0 \Rightarrow) y = -4$

or y -intercept = -4 or $D(0, -4)$

A1cso

3

(b) $3x + 2(1 - 4x) = 7, y = 1 - \frac{4}{3}(7 - 2y)$

elimination of y (or x) (condone one slip)

M1

$$x = -1$$

one coordinate correct

A1

$$y = 5$$

other coordinate correct

coordinates of A(-1, 5)

A1

3

(c) $(5 - 2)^2 + (k + 7)^2 = 5^2$

condone one sign slip within one bracket

M1

(or $k + 7 = 4$ or $k + 7 = -4$)

$$k = -3$$

one correct value of k

A1

or $k = -11$

both correct (and no other values)

A1

3

[11]

Q17.

(a) (i) $(-)(x + 5)^2$

$q = 5$; *condone $(-x - 5)^2$*

M1

$$29 - (x + 5)^2$$

$p = 29$ and $q = 5$

A1

2

(ii) $x = -5$ is line of symmetry

FT $x = -$ 'their q' or correct

B1✓

1

(b) (i) $4 - 10x - x^2 = k(4x - 13)$

$$\Rightarrow x^2 + 4kx + 10x - 13k - 4 = 0$$

Must see both these lines OE

$$\Rightarrow x^2 + 2(2k + 5)x - (13k + 4) = 0$$

AG all correct working and = 0

B1

1

(ii) 2 distinct roots $\Rightarrow b^2 - 4ac > 0$

stated or used (must be > 0)

B1

$$\text{Discriminant} = 4(2k + 5)^2 + 4(13k + 4)$$

condone one slip (may be within formula)

M1

$$4(4k^2 + 20k + 25 + 13k + 4) > 0$$

$$\text{or } 16k^2 + 132k + 116 > 0$$

$$\Rightarrow 4k^2 + 33k + 29 > 0$$

AG > 0 must appear before final line

A1

3

(iii) $(4k + 29)(k + 1)$

correct factors or correct unsimplified quadratic

equation formula
$$\frac{-33 \pm \sqrt{33^2 - 4 \times 4 \times 29}}{8}$$

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M1

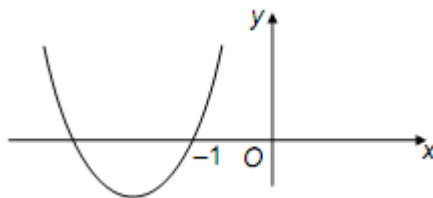
$$k = -\frac{29}{4}, k = -1$$

$$\text{condone } k = -\frac{58}{8}, 7.25 \text{ etc but not left with}$$

square roots etc as above

A1

$$-\frac{29}{4}$$



sketch or sign diagram including values



M1

$$k < -\frac{29}{4}, k > -1$$

condone use of **OR** but not **AND**

A1

Take their final line as their answer

4

[11]

Q18.

(a) $y = \frac{13}{3} - \frac{7}{3}x$

attempt at $y = a + bx$ or $\frac{\Delta y}{\Delta x}$ with 2 correct points

M1

(gradient =) $-\frac{7}{3}$

condone slip in rearranging if gradient is correct

A1

2

(b) (i) $y - 3 = \text{'their grad'}(x - -1)$

or $7x + 3y = k$ and attempt at k using
 $x = -1$ and $y = 3$

or $y = (\text{their } m)x + c$ and attempt at c
using $x = -1$ and $y = 3$

M1

$$y - 3 = -\frac{7}{3}(x + 1) \text{ or } 7x + 3y = 2 \text{ or } y = -\frac{7}{3}x + c, c = \frac{2}{3}$$

*correct equation in any form and replacing
 – – with + sign*

A1cso

2

(ii) (4, -5)

$$x = 4, y = -5$$

withhold if clearly from incorrect working

B1,B1

2

(c) $7x + 3y = 13$ and $3x + 2y = 12$

$\Rightarrow$ equation in x or y only

*must use correct pair of equations and
 attempt to eliminate y (or x)*

M1

$$x = -2$$

A1

$$y = 9$$

A1

3

[9]

Q19.

(a) $(x - 3)^2 + (y + 8)^2$

accept $(y - -8)^2$

B1

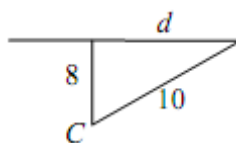
$$= 100$$

condone $RHS = 10^2$ or $k = 10^2$

B1

2

(b) $y = 0 \Rightarrow$ 'their' $(x - a)^2 + b^2 = k$



Alternative

M1

$$(x - 3)^2 = 36 \quad \text{or} \quad x^2 - 6x - 27 (= 0) \quad (\text{PI})$$

A1

$$\Rightarrow x = -3, 9$$

$$(d^2 =) 10^2 - 8^2 \quad \text{M1}$$

$$d^2 = 36 \quad \text{A1} \quad \text{or } d = 6$$

$$\Rightarrow x = -3, 9 \quad \text{A1}$$

A1

3

(c) Line CA has gradient $-\frac{2}{5}$

M1

CA has equation $(y + 8) = -\frac{2}{5}(x - 3)$
any form of correct equation

eg $y = -\frac{2}{5}x + c, \quad c = -\frac{34}{5}$

A1

$$2x + 5y + 34 = 0$$

integer coefficients – all terms on 1 side

A1cso

3

(d) (i) their $(x - 3)^2 + (2x + 1 + 8)^2$ or

$$x^2 + (2x + 1)^2 - 6x + 16(2x + 1) (+ 73)$$

*substituting $y = 2x + 1$ correctly into LHS
 of "their" circle equation and attempt to
 expand in terms of x only*

M1

$$x^2 - 6x + 9 + 4x^2 + 36x + 81 = 100$$

$$\text{or } x^2 + 4x^2 + 4x + 1 - 6x + 32x + 16 + 73 = 100$$

any correct equation (with brackets expanded)

A1

$$\Rightarrow 5x^2 + 30x - 10 = 0$$

must see this line or equivalent

$$\Rightarrow x^2 + 6x - 2 = 0$$

AG; all algebra must be correct

A1cso

3

(ii) $(x + 3)^2 = 11$

or correct use of formula

must get as far as $x = \frac{-6 \pm \sqrt{44}}{2}$

M1

$$x = -3 \pm \sqrt{11}$$

exactly this

A1cso

2

[13]

Q20.

(a)

For either 6 or $6x^0$

B1

$$\frac{dy}{dx} = 6 - 3x^{\frac{1}{2}}$$

$$Ax^{\frac{3}{2}-1}, A \neq 0 \text{ OE}$$

M1

$6 - 3x^{\frac{1}{2}}$ or $6 - 3\sqrt{x}$ with no '+c'

[If unsimplified here, A1 can be awarded retrospectively if correct simplified expression is seen explicitly in (b)(i).]

A1

3

(b) (i) $6 - 3x^{\frac{1}{2}} = 0$

*Equating c's $\frac{dy}{dx}$ to 0 PI by correct ft
rearrangement of c's $dy/dx = 0$*

M1

$$x^{\frac{1}{2}} = 2 \Rightarrow x = 2^2$$

$$x^{\frac{1}{2}} = k \quad (k > 0), \text{ to } x = k^2.$$

PI by correct value of x if no error seen

m1

$M(4, 8)$

SC If M0 award B1 for (4, 8)

A1

3

(ii) Eqn of normal at M is $x = 4$

Ft on $x = c$'s x_M

B1F

1

(c) (i) When $x = \frac{9}{4}, \frac{dy}{dx} = 6 - 3 \times \frac{3}{2} = \frac{3}{2}$

Attempt to find $\frac{dy}{dx}$ when $x = \frac{9}{4}$

M1

Gradient of normal at $P = -\frac{2}{3}$

$m \times m' = -1$ used

m1

Eqn of normal: $y - \frac{27}{4} = -\frac{2}{3} \left(x - \frac{9}{4} \right)$

ACF eg $y = -\frac{2}{3}x + \frac{33}{4}$

A1

$$12y - 81 = -8x + 18 \Rightarrow 8x + 12y = 99$$

Coeffs and constant must now be positive integers, but accept different order

eg $12y + 8x = 99$

A1

(ii) $8(4) + 12y = 99$

*Solving c's answer (b)(ii), (must be in form $x = \text{positive const}$), with c's answer (c)(i).
 PI by correct earlier work and correct coordinates for R.*

M1

$$R\left(4, \frac{67}{12}\right)$$

Accept 5.58 or better as equivalent to $\frac{67}{12}$

A1

2

[13]

Q21.

(a) $y = \frac{14}{3} - \frac{2}{3}x$

Attempt at $y = \dots$

M1

Gradient $AB = -\frac{2}{3}$

Condone error in rearranging equation

A1

2

(b) (i) $y - 7 = \text{"their grad } AB" (x - 3)$

*or $2x + 3y = k$ and sub $x = 3, y = 7$
 or $y = mx + c, m = \text{their grad } AB$ and
 attempt to find c using $x = 3, y = 7$*

M1

$$y - 7 = -\frac{2}{3}(x - 3)$$

OE

$$2x + 3y = 27, \quad y = -\frac{2}{3}x + 9 \text{ etc}$$

A1

2

- (ii) $m_1 m_2 = -1$
or negative reciprocal (stated or used PI)

M1

$$\Rightarrow \text{grad } AD = \frac{3}{2}$$

FT their grad AB

A1ft

$$y - 7 = \frac{3}{2}(x - 3)$$

Any correct equation unsimplified

A1

$$\Rightarrow 3x - 2y + 5 = 0$$

*Integer coefficients; all terms on one side,
 condone different order or multiples.
 eg $0 = 4y - 6x - 10$*

A1

4

- (c) $2x + 3y = 14$ and $5y - x = 6$ used
with x or y eliminated (generous)
 $2(5y - 6) + 3y = 14$ etc

M1

$$x = 4,$$

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A1

$$y = 2$$

B(4, 2) full marks NMS

A1

3

[11]

Q22.

(a) $\{S_{40} = \frac{40}{2}[2a + (40 - 1)d]\}$

M1

$$20(2a + 39d) = 1250$$

A1

$$\{25^{\text{th}} \text{ term} \Rightarrow a + (25 - 1)d$$

M1

$$a + 24d = 38$$

A1

Dep on both previous two Ms. Solving two equations in a and d simultaneously

m1

$$18d = 27 \Rightarrow d = 1.5$$

AG Be convinced

SC Using the given answer for d : mark out of a maximum of 4/6 as

M1A1M1A1 {conclusion also needed in last A mark} (m0A0)

A1cso

6

$$(b) \quad a = 38 - 24 \times 1.5 \\ = 2$$

PI if using $a = 2$ in (b)

If using eg $a = 38$ award this M mark at

$$\text{stage: no. of terms} \quad \frac{100 - 38}{1.5} + 1 + 24$$

M1

$$a + (n - 1) 1.5 < 100$$

M1

$$n < \frac{100 - a}{1.5} + 1$$

$$n < 66.333 \dots$$

$\Rightarrow$ number of terms < 100 is 66

NMS mark as B3 for 66 else B0

A1

3

[9]

Q23.

(a) (i) $y = -\frac{3}{5}x + \frac{11}{5}$

Or correct expression for gradient using two correct points

Attempt at $y = f(x)$

M1

Or answer $= \frac{3}{5}$ or $-\frac{3}{5}x$ gets M1

But answer of $\frac{3}{5}x$ gets M0

(Gradient of $AB = -\frac{3}{5}$)

Correct answer scores 2 marks.

Condone error in rearranging formula if answer for gradient is correct.

A1

2

(ii) $m_1 m_2 = -1$

Used or stated

M1

Gradient of perpendicular $= \frac{5}{3}$

ft their answer from (a) (i) or correct

A1ft

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$y - 1 = \frac{5}{3}(x - 2)$

OE

$5x - 3y = 7$; or $y = \frac{5}{3}x + c$, $c = -\frac{7}{3}$ etc

CSO

A1

3

(b) Eliminating x or y but must use

$3x + 5y = 11$ & $2x + 3y = 8$

An equation in x only or y only

M1

$x = 7$

A1

$$y = -2$$

A1

Answer only of (7, -2) scores 3 marks

3

[8]

Q24.

(a) $k(x^2 + 3) = 2x + 2$

$$\Rightarrow kx^2 - 2x + 3k - 2 = 0$$

AG OE all terms on one side and = 0

B1

1

(b) (i) Discriminant = $(-2)^2 - 4k(3k - 2)$

Condone one slip (including x is one slip)

Condone 2² or 4 as first term

M1

$$= 4 - 12k^2 + 8k$$

condone recovery from missing brackets

A1

Two distinct real roots $\Rightarrow b^2 - 4ac > 0$

$$4 - 12k^2 + 8k > 0$$

"their discriminant in terms of k" > 0

Not simply the statement $b^2 - 4ac > 0$

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B1ft

$$\Rightarrow 12k^2 - 8k - 4 < 0$$

Change from > 0 to < 0 and divide by 4

$$\Rightarrow 3k^2 - 2k - 1 < 0$$

AG CSO

A1

4

(ii) $(3k + 1)(k - 1)$

Correct factors or correct use of formula

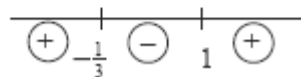
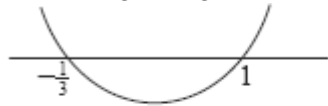
M1

Critical values 1 and $-\frac{1}{3}$

May score M1, A1 for correct critical values seen as part of incorrect final answer with or without working

A1

Use of sign diagram or sketch



If previous A1 earned, sign diagram or sketch must be correct for M1

M1

Otherwise, M1 may be earned for an attempt at the sketch or sign diagram using their critical values.

$$\Rightarrow -\frac{1}{3} < k < 1 \quad \text{or} \quad 1 > k > -\frac{1}{3}$$

*Full marks for correct final answer with or without working
≤ loses final A mark*

A1

condone $-\frac{1}{3} < k$ AND $k < 1$ for full marks but not OR or “,” instead of AND

Answer only of $1 < k < -\frac{1}{3}$ or

$k < -\frac{1}{3}; k < 1$ etc scores M1, A1, M0 since the correct critical values are evident

Answer only of $\frac{1}{3} < k < 1$ etc where critical values are not both correct gets M0, M0

Q25.

(a) $\frac{dy}{dx} = \frac{45}{2}x^{\frac{1}{2}} - \frac{5}{2}x^{\frac{3}{2}}$

One power correctly obtained

M1

A1 for each term on the RHS coeffs simplified

A2,1,0

3

(b) $\frac{45}{2}x^{\frac{1}{2}} - \frac{5}{2}x^{\frac{3}{2}} = 0$

can't's (a) = 0

M1

$$\frac{5}{2}x^{\frac{1}{2}}(9 - x) = 0$$

Must be solving eqn of form $ax^m + bx^n = 0$, m and n non-zero, with at least one of m and n non-integer and reaching a stage from which the non-zero value of x can be stated. Must deal with powers of x correctly and any squaring of kx^p terms or expressions must be correct.

m1

At M , $x = 9$

A1

$y_M = 162$

M1 must be scored, else 0/4

A1

4

(c) At $P(1, 14), \frac{dy}{dx} = \frac{45}{2} - \frac{5}{2} = 20$

Attempt to find $y'(1)$

M1

Tangent at P : $y - 14 = m(x - 1)$

$m =$ can't's value of $y'(1)$

m1

$$y - 14 = 20x - 20; y = 20x - 6$$

CSO; AG

A1

3

- (d) Tangent at M : $y = 162$
ft $y = \text{cand's } y_M$

B1F

At R , $162 = 20x - 6; x = 8.4$

Solving cand's numerical $y_M = 20x - 6$ to find a value for x

M1

Distance $RM = |x_M - x_R| = 9 - 8.4 = 0.6$

ft on coordinates of M

A1F

3

[13]

Q26.

(a) $x^2 + (y - 5)^2$
 $b = 5$

B1

RHS = 5

$k = 5$

B1

2

- (b) (i) Centre (0, 5)
FT their b from part (a)

B1ft

1

(ii) Radius = $\sqrt{5}$

FT their k from part (a); RHS must be > 0

B1ft

1

- (c) (i) $x^2 + 4x^2 - 20x + 20 = 0$
May substitute into original or "their (a)"

M1

$$\Rightarrow x^2 - 4x + 4 = 0$$

CSO; AG

A1

2

(ii) $(x - 2)^2 = 0$ or $x = 2$

M1

Repeated root implies tangent

Or $b^2 - 4ac$ shown = 0 plus statement

E1

Point of contact is $P(2, 4)$

A1

3

(d) $(CQ^2 =) 1^2 + 1^2$

FT their C

M1

$$\sqrt{2} < \sqrt{5} \Rightarrow Q \text{ lies inside circle}$$

CQ or CQ^2 OE must appear for A1

A1CSO

2

[11]

Q27. EXAM PAPERS PRACTICE

(a) $x^2 + 7 = k(3x + 1) \Rightarrow x^2 - 3kx + 7 - k = 0$

AG

B1

1

(b) $b^2 - 4ac = (-3k)^2 - 4(7 - k)$

Clear attempt at $b^2 - 4ac$

Condone slip in one term of expression

M1

(2 distinct roots when) $b^2 - 4ac > 0$

Must involve k

B1

$$9k^2 + 4k - 28 > 0$$

CSO; AG

A1

3

(c) $(9k - 14)(k + 2)$

Factors or formula correct unsimplified

M1

Critical points -2 and $\frac{14}{9}$

A1

Sketch $\cup$ or sign diagram correct

+ve	-ve	+ve
	-2	$\frac{14}{9}$

M1

$$k < -2, k > \frac{14}{9}$$

A1

4

[8]

Q28.

- (a) *L*: straight line with positive gradient and negative intercept on *y*-axis

Line must cross both axes but need not reach the curve

B1

cutting at $(\frac{1}{3}, 0)$ and $(0, -1)$
(intercepts stated or marked on sketch)

Condone 0.33 or better for $\frac{1}{3}$

B1

C: attempt at parabola $\cup$ or $\cap$

through $(-3, 0)$ and $(1, 0)$ or
values -3 and 1 stated as intercepts
on x -axis

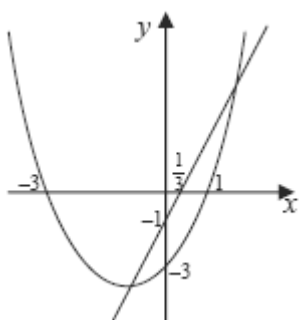
B1

∪ shaped graph – vertex below x -axis and cutting x -axis twice

M1

through $(0, -3)$ and minimum point to left of y -axis

A1



(y-intercept or coordinates marked)

5

(b) $(x + 3)(x - 1) = 3x - 1$

M1

$$x^2 + 3x - x - 3 - 3x + 1 = 0$$

$$\Rightarrow x^2 - x - 2 = 0$$

AG; must have “= 0” and no errors

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A1

2

(c) $(x - 2)(x + 1) = 0$

$(x \pm 1) (x \pm 2)$ or use of formula (one slip)

M1

$$\Rightarrow x = 2, -1$$

correct values imply M1A1

A1

Substitute one value of x to find y

m1

Points of intersection $(2, 5)$ and $(-1, -4)$

May say $x = 2, y = 5$ etc

SC: $(2, 5) \Rightarrow B2$

$(-1, -4) \Rightarrow B2$ without working

A1

4

[11]

Q29.

(a) $V = x(9 - 3x)^2$

Attempt at V in terms of x (condone slip
when rearranging formula for $y = 9 - 3x$)

or $(9 - 3x)^2 = 81 - 54x + 9x^2$

M1

$$V = x(81 - 54x + 9x^2)$$

$$= 81x - 54x^2 + 9x^3$$

AG; no errors in algebra

A1

2

(b) (i) $\frac{dV}{dx} = 81 - 108x + 27x^2$

One term correct

M1

Another correct

A1

All correct (no + c etc)

A1

$$= 27(x^2 - 4x + 3)$$

CSO; all algebra and differentiation correct

A1

4

(ii) $(x - 3)(x - 1)$ or $(27x - 81)(x - 1)$ etc

"Correct" factors or correct use of formula

M1

$$\Rightarrow x = 1, 3$$

SC: B1, B1 For $x = 1, x = 3$ found by
inspection (provided no other values)

A1

2

(c) $\frac{d^2V}{dx^2} = -108 + 54x$ (condone one slip)

ft their $\frac{dV}{dx}$ (may have cancelled 27 etc)

M1

CSO; all differentiation correct

A1

2

(d) (i) $x = 3 \Rightarrow \frac{d^2V}{dx^2} = 54; \quad x = 1 \Rightarrow \frac{d^2V}{dx^2} = -54$

ft their $\frac{d^2V}{dx^2}$ and their **two** x -values

B1ft

1

(ii) ($x =$) 1 (gives maximum value)

Provided their $\frac{d^2V}{dx^2} < 0$

E1

1

(iii) $V_{\max} = 36$
CAO

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B1

1

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[13]

Q30.

(a) (i) $y = -\frac{3}{5}x + \dots; \quad \text{Gradient } AB = -\frac{3}{5}$

Attempt to find $y =$ or $\Delta y / \Delta x$ or $\frac{3}{5}$ or $3x/5$

M1

Gradient correct – condone slip in $y = \dots$

A1

(ii) $m_1 m_2 = -1$

Stated or used correctly

M1

Gradient of perpendicular = $\frac{5}{3}$
ft gradient of AB

A1 ✓

$$\Rightarrow y + 2 = \frac{5}{3}(x - 6)$$

CSO Any correct form eg $y = \frac{5}{3}x - 12$,
 $5x - 3y = 36$ etc

A1

3

(b) Eliminating x or y (unsimplified)

Must use $3x + 5y = 8$; $2x + 3y = 3$

M1

$$x = -9$$

A1

$$y = 7$$

$B(-9, 7)$

A1

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3

(c) $4^2 + (k + 2)^2 (=25)$ or $16 + d^2 = 25$

Diagram with 3, 4, 5 triangle

M1

$$k = 1$$

A1

Condone slip in one term (or $k + 2 = 3$)

or $k = -5$

SC1 with no working for spotting one correct value of k . Full marks if both

values spotted with no contradictory work

A1

3

[11]

Q31.

(a) $(x + 1)^2 + (y - 6)^2$

B1 for one term correct or missing + sign

B2

$(1 + 36 - 12 = 25) \text{ RHS} = 5^2$

Condone 25

B1

3

(b) (i) Centre $(-1, 6)$

FT their a and b from part (a) or correct

B1 ✓

1

(ii) Radius = 5

FT their r from part (a) RHS must be > 0

B1 ✓

1

(c) Attempt to solve "their" $x^2 + 2x + 12 = 0$

Or comparing "their" $y_c = 6$ and their $r = 5$

M1

(all working correct) so no real roots
or statement that does not intersect

may use a diagram with values shown

$$\begin{cases} r < y_c \text{ so does not intersect} \\ \text{condone } \pm 1 \text{ or } \pm 6 \text{ in centre for A1} \end{cases}$$

A1

2

(d) (i) $(4 - x)^2 = 16 - 8x + x^2$

Or $(-2 - x)^2 = 4 + 4x + x^2$

B1

$x^2 + (4 - x)^2 + 2x - 12(4 - x) + 12 = 0$

or $(x + 1)^2 + (-2 - x)^2 = 25$

*Sub $y = 4 - x$ in circle eqn (condone slip)
or "their" circle equation*

M1

$$\Rightarrow 2x^2 + 6x - 20 = 0 \Rightarrow x^2 + 3x - 10 = 0$$

AG CSO (must have = 0)

A1

3

(ii) $(x + 5)(x - 2) = 0 \Rightarrow x = -5, x = 2$

*Correct factors or unsimplified solution to
quadratic (give credit if factorised in part (i))*

M1

Q has coordinates $(-5, 9)$

SC2 if Q correct. Allow $x = -5$ $y = 9$

A1

2

(iii) Mid point of 'their' $(-5, 9)$ and $(2, 2)$

*Arithmetic mean of **either** x or y coords*

M1

$$\left(-1\frac{1}{2}, 5\frac{1}{2}\right)$$

Must follow from correct value in (ii)

A1

2

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[14]

Q32.

(a) (i) $(x + 5)^2$

$$p = 5$$

B1

$$-6$$

$$q = -6$$

B1

2

(ii) $x_{\text{vertex}} = -5$ (or their $-p$)

may differentiate but must have $x = -5$

B1 ✓

$y_{\text{vertex}} = -6$ (or their q)
and $y = -6$. Vertex $(-5, -6)$

B1 ✓

2

(iii) $x = -5$

B1

1

(iv) Translation (not shift, move etc)
and NO other transformation stated

E1

through $\begin{bmatrix} -5 \\ -6 \end{bmatrix}$ (or 5 left, 6 down)
either component correct

M1

M1, A1 independent of E mark

A1

3

(b) $x+11 = x^2 + 10x + 19$
quadratic with all terms on one side of equation

$\Rightarrow x^2 + 9x + 8 = 0$ or $y^2 - 13y + 30 = 0$

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$(x+8)(x+1) = 0$ or $(y-3)(y-10) = 0$
attempt at formula (1 slip) or to factorise

ml

$\left. \begin{matrix} x = -1 \\ y = 10 \end{matrix} \right\}$ or $\left. \begin{matrix} x = -8 \\ y = 3 \end{matrix} \right\}$

both x values correct

A1

both y values correct and linked

SC $(-1, 10)$ **B2**, $(-8, 3)$ **B2** no working

A1

4

[12]

Q33.

(a) (i) $(x - 2)^2$
 $p = 2$

B1

$+ 5$
 $q = 5$

B1

2

- (ii) Minimum point $(2, 5)$ or $x = 2, y = 5$
B1 for each coordinate correct or ft
Alt method M1, A1 sketch, differentiation

B2 ✓

2

(b) (i) $12 - 2x = x^2 - 4x + 9$
 Or $x^2 - 4x + 9 + 2x = 12$
 $\Rightarrow x^2 - 2x - 3 = 0$
AG (be convinced) (must have = 0)

B1

1

(ii) $(x - 3)(x + 1) = 0$
Attempt at factors or quadratic formula or one value spotted

M1

$x = 3, -1$
Both values correct & simplified

A1

Substitute one value of x to find y
May substitute into equation for L or C

M1

Points are (3, 6) and (-1, 14)
y-coordinates correct linked to x values

A1

4

[9]

Q34.

(a) $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} - 3$

One term correct

M1

Both correct

A1

2

(b) (i) When $x = 0$, $\frac{dy}{dx} = -3$

Ft provided answer < 0.

B1F ✓

Eqn of tangent at O is $y = -3x$
OE Ft on $y'(0)$

B1F ✓

2

(ii) At (9,0) $\frac{dy}{dx} = \frac{3}{2}(9)^{\frac{1}{2}} - 3$

Attempt to find $y'(9)$

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Eqn tangent at A is $y - 0 = y'(9)[x - 9]$
OE

m1

$$\Rightarrow y = \frac{3}{2}(x - 9) \Rightarrow 2y = 3x - 27$$

CSO. AG

A1

3

(iii) Eliminating $y \Rightarrow -6x = 3x - 27$

*OE method to one variable
(eg $2y = -y - 27$)*

M1

$$9x = 27 \Rightarrow x = 3$$

*[A1F for each coordinate; only ft on $y = kx$ tangent
in (b)(i) for $k < 0$]*

A1F

$$\text{When } x = 3, y = -9 \{ P(3, -9) \}$$

A1F

3

(c) $\int \left(x^{\frac{3}{2}} - 3x \right) dx = \frac{2}{5} x^{\frac{5}{2}} - \frac{3x^2}{2} (+c)$
One power correct

M1

Condone absence of "+ c" and unsimplified forms

A2,1,0

3

(d) $\int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx =$
PI

B1

$$= \frac{2}{5} \times 9^{\frac{5}{2}} - \frac{3}{2} \times 9^2 - 0$$

Correct use of limits following integration

M1

$$= -24.3$$

$$\text{Area of triangle } OPA = \frac{1}{2} \times 9 \times |y_P|$$

M1

$$\text{Sh.Area} = \frac{1}{2} \times 9 \times |y_P| - \left| \int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx \right|$$

OE

M1

$$= 40.5 - 24.3 = 16.2$$

A1

5

[18]

Q35.

(a) (i) Gradient $AB = \frac{1-7}{5-1}$

Must be y on top and subtraction of coordinates

M1

$$= -\frac{6}{4} = -\frac{3}{2} = -1.5$$

Any correct equivalent

A1

2

(ii) $y-7 = m(x-1)$ or $y-1 = m(x-5)$

Verifying 2 points or $y = -\frac{3}{2}x + c$

M1

leading to $3x + 2y = 17$

AG (or grad & 1 point verified)

A1

2

(b) Attempt to eliminate x or y : $7x = 42$ etc

Solving $x - 4y = 8$; $3x + 2y = 17$

M1

$$x = 6$$

A1

$$y = -\frac{1}{2}$$

C is point $(6, -\frac{1}{2})$

A1

3

(c) Grad of perp = $-1 /$ *their* gradient AB

Or $m_1 m_2 = -1$ used or stated

M1

$$= \frac{2}{3}$$

ft their gradient AB

A1 ✓

$$y - 7 = \frac{2}{3}(x - 1) \quad \text{or } 3y - 2x = 19$$

CSO Any correct form of equation

A1

3

[10]